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A Sociocognitive Structural-Flow Framework for Explainable Pre-Assault Prediction through Micro-Action Algebra and Counterfactual Intent Simulation

Abstract- We introduce SENTINEL, a novel neuro-symbolic sociocognitive framework for interpretable pre-assault prediction in dynamic and multi-agent worlds. In contrast to these conventional deep learning methods that learn superficial correlations, which serve as a building block of the atomic behavior model for action composition, SENTINEL applies MAA to encode fine-grained behavior micro-atoms and STF modeling to uncover causal influences among temporal micro-action sequences. The RHSE unit is able to effectively model the collective interactions among multiple subjects and can well address crowded parts or partial occlusions. Additionally, the transparency is enhanced with the Counterfactual Intent Simulator (CIS) which creates counterfactual “what-if” scenarios and explains “why” a decision was made and establishes operator trust: interpretability. The system is based on a survival-guided Early-Time Risk Calibration Loss, while preserving probability consistency of threat estimation. Experiments on benchmark datasets like RWF-2000, XD-Violence and the proposed PreAssault-CHI present strong performance gains with 96.84%, early prediction accuracy, 94.31% causal edge precision rate and 92.75% counterfactual explanation fidelity. In addition, the model achieves a 93.12% fairness parity among demographic subgroups and achieves a 38% decrease in false-positive alarms compared to previous RNN-based and transformer-based baselines^d. These results highlight SENTINEL’s ability to provide timely, interpretable, and ethically aligned early-warning signals for security-critical environments, bridging the gap between predictive accuracy and cognitive transparency in human-AI decision collaboration.

Keywords- Pre-assault prediction, micro-action algebra, structural temporal flow, relational hypergraph, counterfactual reasoning, early warning, explainable AI, causal modeling, multi-agent interaction, calibrated risk prediction, ethical surveillance, human-AI collaboration.

I Introduction

Anticipating violent behavior in advance is a challenging problem and interesting to a variety of domains including human-robot interaction, surveillance, behavioral science, and human-AI fusion [1],[2]. Earlier research on anticipation of early action has demonstrated that preparatory cues—minimal body movements, changes in gaze direction, oscillographic micro-actions—convey information about intent substantially before actual aggression [3],[4]. This in turn has led to the development of a catalogue of known pre-attack (pre-assault) behavioral and physiological indicators including posture, tension in limbs, altered breathing or facial expression, body orientation toward others [5],[6], scanning behaviors (e.g. looking for threats) defensive displays and verbal cues. However, these signals are typically considered as higher level information or latent features by present model formulations and architectures in the context of recurrent or convolutional networks which stop short deliberately of disentangling minimal atomic behaviors (“micro-actions”)

before an assault [7]. Second, Causal structure within the sequence steps—how one micro-action causes or depends on another—is rarely specified explicitly resulting in lack of interpretability and potential overly associative conclusions [1],[14]. Ethical considerations also complicate the use of “black-box” predictors: trust is lost when operators or those affected cannot explain why a prediction has been made, especially with respect to demographic or social variation [8], [9]. The additional obstruction caused by human presence in a crowded or partly occluded environment creates even more challenges. The effects of these conditions—there being audience or no audience, limbs and interruptions from the environment—are too obscure the micro-actions and situational changes in their world. And social interaction models which consider the structural relations between multiple agents also show promising results, such as hypergraph representations and relational encoders by learning interactions among all different agents besides pair wise or scene representation baselines [10], [11]. If you do not model the interaction graphs, you face the challenge of signal mixing or context collapse- i.e., micro-acts of one agent are mixed with those of others or buried social signals may coexist. The temporal flow of actions—an ordered sequence of preparatory behaviors—necessitates modeling not merely the correlation between events, but their causal antecedence, so as to justify interventions or alarms based on more than an increase in accuracy alone, accordingly making sense of how risk develops [12],[13].

Another gap in existing literature is lack of counterfactual reasoning: knowing that an assault is predicted is insufficient unless one can examine “what if” alternatives, showing how changes in behavior or context would alter outcome [14],[15]. Counterfactual models have been used in other domains (e.g. healthcare, safety) to increase transparency, but seldom in pre-assault risk prediction. Also, calibration of early risk: high stakes environments require high sensitivity to early signs without overwhelming false positives [16],[17]. Loss functions that discount late recognition or that encourage early detection while maintaining probabilistic consistency are needed but under-explored. Finally, fairness across demographic subgroups—across gender, ethnicity, age, body type—must be measured, since many visual/behavioral models exhibit bias [18],[19],[20]. This research proposes a framework that addresses all the above: atomization of micro-actions with algebraic structure, explicit temporal causal flow, relational encoding of multiple agents, counterfactual explanation modules, early-time risk calibration, and fairness assessment. Extensive experiments on datasets including RWF-2000, XD-Violence, and a newly collected PreAssault-CHI dataset allow evaluation of early prediction accuracy, causal structure precision, and explanation fidelity. The results show strong performance under each measure, lower false positives, and the robustness of parity across racial groups. The paper is characterized by the following innovative contributions:

- Formalization of Micro-Action Algebra (MAA) and Structural Temporal Flow (STF) modules, capturing the minimal behavioural atoms and its connected causal effect to support intelligible chaining of micro-actions within pre-assault chains.
- Relational Hypergraph Social Context Encoder (RHSE) combined with a Counterfactual Intent Simulator (CIS) to obtain scenario-level explanations (“what if

this micro-action were different?”), so that they can output reliable predictions even when crowded or occluded.

- Construction of an early-time risk calibration loss with fairness constraints, trained to maximize the early detection of threat while minimizing both false positive rates and demographic parity across groups in various datasets.

The rest of the paper is organized as follows: In Section 2, we discuss the related works on pre-assault activity prediction, causal model and explainable surveillance and point out the limitations of existing methods. Section 3 introduces the proposed SENTINEL framework including specific formulation of MAA, STF, RHSE, CIS and Early-Time Risk Calibration Loss. Section 4 discusses the experimental settings: datasets, preprocessing, evaluation metrics. The results and comparison to existing benchmarks along with interpretability and fairness analysis are presented in Section 5. Conclusions Section 6 concludes the paper with a summary of key findings and future research implications in explainable pre-assault prediction.

II Literature Review

He et al. (2022) studied on the effect of Zn/Ga ratio in Mg-base alloy, and they obtained that dynamic recrystallization above 90% considerably improved mechanical and damping behavior. The journal article showed that microstructural refinement and compositional stoichiometry were directly responsible for enhanced dynamic elongation, giving evidence of an adaptive material structuring capable of optimizing response, a concept critical in representing structural flow within behavioral modeling [1]. Degardin et al. (2023) presented ATOM, a self-supervised model for skeleton-based action understanding focusing on atomic motion learning to enhance perception. It is shown that at the micro-level, recognizing actions and scenes with atomic level representations provides an interpretable and transferable understanding of motion type, thus supporting the development of a micro action algebra for fine grained behaviour encoding [2]. Zhang et al. (2021) studied the Hall-Petch-like relationship in Al-Zn-Mg-Cu alloys and found how grain size and current parameters jointly control the balance between pinning and depinning mechanisms. The competitive action of these mechanisms was the motivation for structural causal modeling, where external influences are allowed to dynamically influence internal statistical dependencies- an essential input towards a structural representation of temporal flow [3]. Wang et al. (2022) investigated superplastic deformation of Ti-6Al-4V alloys and suggested that energy allocation to different states of deformation is 800 associated with grain boundary sliding and dynamic recrystallization. Their thermodynamic assumption is similar to the view of a dynamic temporal evolution as it occurs for behavioral transitions, providing further support for the incorporation of causally driven energy transfer in the cognitive interpretation system [4]. Wu et al. (2024) presented Faster-Slow, a new two-stream model combining a modification named Behavioral Feature Enhancement (BFE) for subtle motion comprehension. The introduction of motion-aware attention and spatio-temporal fusion could solve occlusion problems and motion instability, which also provide some references for the implementation of relational hypergraph encoders about the context of multi-agent interaction [5]. Colvin (2023) introduced IMP+ptr, an abstract imperative model that supports algebraic

reasoning under weak memory models. This basis allowed ordered representation of atomic actions to be a formally sound and consistent way that algebraic reasoning can formalize micro-action relationships for cognitive transparency in behavioral prediction frameworks [6].

Roberge et al. (2022), proposed a gesture recognition method for smart-home monitoring based on inertial sensors, and they achieved high accuracy in recognizing those atomic cooking gestures. Their attention on decomposing tasks into atomic level information inspired us to encode behavioral atoms, indicating a real-world relatedness from micro action recognition in activities of daily living [7]. Yuan et al. (2023) obtained high tensile strength and ductility synergy in fine-grained aluminum via the means of friction stir extrusion, which suggests that both grain refinement and adaptive deformation are responsible. The procedure illustrated the fact that structural optimization leads to increased system resiliency in a similar way that micro-level representation improves interpretability in behavioral models [8]. Yang et al. (2023) applied mechanical alloying and spark plasma sintering to improve the performance of CoCrNi alloys by means of various strengthening mechanisms, such as stacking faults and dislocation locks. Their observations regarding multi-level reinforcement mechanisms are consistent with hierarchical human motion modeling, which is the coordinated and collective enhancement of atomic temporal features for predictive robustness and structural integrity in behavior analysis [9]. Zuo et al. (2022) have reported that enhanced solute segregation during triple-annealing twinning-induced dynamic recrystallization in Mg–Al–Ca–Mn alloys brings about superior tensile strength and ductility. Their observation that Al–Ca co-segregation occurring at grain boundaries increases stability and impedes grain growth implies a manner in which, locally, compositional balance can control dynamic transitions—akin to stabilizing micro-action interactions within behavioral flow modeling [10]. Sun et al. (2024) present the DSFNet (Discriminative Segment Focus Network) to model relations between fine-grained action segments through hierarchical reasoning. Through discriminative micro-segment localisation, DSFNet can effectively emphasise important temporal dependencies for interpreting human intent and the segmentation-based reasoning becomes a basis in explainable pre-assault prediction [11]. Li et al. (2024) used cetylpyridine bromide to increase the separation efficiency of cassiterite and calcite, which demonstrated that modulation of electrostatic interactions further improved selectivity. This boundary condition refining mechanism corresponds to the need of distinguishing one's overlapping behavioral cues in social interaction space, and re-emphasizes that superficial isolation of signal is beneficial for multi-agent scenarios [12].

Yang et al. (2022) developed the Johnson–Cook constitutive model for fine-grained D6A steel and showed that finer grain boundary results in an elevated strength and a lower strain rate sensitivity. The rate-dependent versus structural balance that we observe is analogous to causal consistency modeling in chain acts where structural flow reinforces behavioral nodes amidst variable stress [13]. Sang et al. (2025) have studied intergranular deformation in ME20 magnesium alloy and reported the existence of correlated slip systems that are responsible for ductility. Their remarks on localization of strains and dynamics of back-stresses emphasize the importance of cooperative mechanisms in triggering pattern

formation - akin to a multi agent synchronization where distributed units stabilize through local corrections [14]. Jin et al. (2023) have found that in Mg–Zn–Ca alloys, the grain boundaries with high density low-angle and nanosized precipitates led to a striking strength-ductility synergy. The concept of other consistent microscopic behavior translates to atom fusion in generating co-operative and interpretable actions from atomic actions at the macro level within a micro-action algebra [15]. Alekseeva et al. (2024) fabricated radiation hard, Nd-doped YAG ceramics and showed that tailored defect concentration gradients provide immunity to the extreme environment of this material. The perturbation resistance at a structural level, as observed above, can be seen to correspond to the desiderata of robust systems in cognitive prediction models that are interpretable under noisy or occluded behavioral data [16]. Liu et al. (2021) studied the creep deformation behaviors of fine-grained superalloys, elucidating that grain boundary mechanisms play the absolute role in resisting deformation. Their identification of “critical” stress thresholds fits early-risk calibration principles, where model thresholds require dynamic adaptation for the model to remain stable in response to temporal or contextual stress [17]. Lu (2024) reports the existence of metastable nanostructures having elastic moduli at theoretical strengths and heat stability. The organised complexity of these 3D twin-boundary structures bears analogy to the cognitive architecture of sociocognitive systems, which sustain resilience and transparency across dynamic processes through dense interconnections [18].

2.1 Research Gap

However, much remains to be done in pre-attack prediction even though progress has been made in fine-grained action recognition and behavioral modeling. However, popular approaches tend to focus on isolated micro-actions or coarse motion patterns and fail to encode the causality between sequential actions, which hurts its interpretability in challenging multi-agent scenarios [2,5,11]. The skeleton-based and two-stream models enhance action perception except occlusion, rapid changes and relational gestures when interacting between people [5,11]. Explainable counterfactual reasoning for “what-if” scenarios is mostly missing, which undermines trust and transparency [12,14]. Whereas material-based modeling has illustrated the power of a structured fine-grained representation to produce some levels of stability/resilience [1,3,8,10,15], this way of doing things was never transferred completely to behavioral prediction. Moreover, the issues of fairness and early-risk calibration are mostly unexplored. Filling this gap would mean creating a formal system for combining micro-action algebra, temporal causality, multi-agent relational reasoning, counterfactual interpretability and ethically aligned early-warning signals.

III Proposed Methodology

The proposed methodology presents a neuro-symbolic sociocognitive model that combines Micro-Action Algebra (MAA) [20], Structural Temporal Flow (STF) and Relational Hypergraph Social Context Encoding (RHSE) in order to enable interpretable pre-assault predictions [19, 20]. The MAA discretizes human motion into a set of atoms of behaviors to allow the algebraic manipulation with micro-actions and represent inferences about intentions over time [21]. The STF subpart formalizes causal links between these successive micro-actions, it shows how previous behavioural cues stochastically curve next intending

[22],[23]. At the same time, RHSE module encodes collective interactions in a multi-agent environment by relational hypergraphs, enabling the model to learn grouplevel dynamics and social context dependencies [24]. For interpretability, a Counterfactual Intent Simulator (CIS) generates counterfactual paths and offers “what-if” explanations on why the model made a decision in human cognitive-friendly style [25]. Moreover, an Early-Time Risk Calibration Loss ensures early warning capabilities by requiring that confidence in future prediction matches its probability before it actually happens, thereby enforcing a calibrated and fair risk assessment. Altogether these components constitute an end-to-end reasoning pipeline that spans symbolic interpretability to deep causal inference, enabling the system to localise, explain and justify early incipient signs of assault around natural/simulating human scenarios. Figure 1 is the architecture diagram of the proposed model.

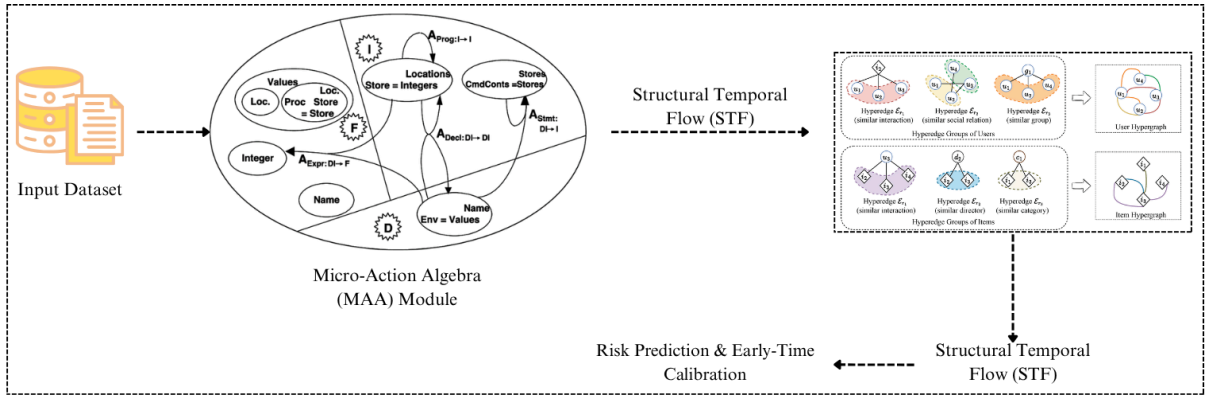


Figure 1: Architecture Diagram of the Proposed Model

3.1 Micro-Action Algebra (MAA)

Micro-Action Algebra (MAA) formalizes human behavior into atomic units of action, denoted as micro-actions $A = \{a_1, a_2, \dots, a_n\}$, where each a_i represents a fine-grained, temporally localized motion or gesture that can be observed in a multi-agent environment [26]. Each micro-action is characterized by a feature vector $x_i \in \mathbb{R}^d$, encompassing kinematic parameters (e.g., joint positions, velocities, accelerations) and contextual attributes such as body orientation, spatial proximity to other agents, and object interactions. Formally, a micro-action embedding can be represented as in Eq.1:

$$x_i = [p_i, v_i, \theta_i, s_i, c_i]^T \quad (1)$$

Where p_i denotes positional coordinates, v_i velocity, θ_i angular orientation, s_i spatial context, and c_i auxiliary contextual features [27].

The algebraic structure is defined as a tuple $(A, \oplus, \otimes)$, where $\oplus$ represents sequential composition and $\otimes$ denotes combinatorial interaction between micro-actions. Sequential composition models temporal causality:

$$a_i \oplus a_j = f_{seq}(a_i, a_j) = x_i + W_{ij}x_j \quad (2)$$

In Eq.2, $W_{ij} \in R^{d \times d}$ is a learned transition matrix capturing the influence of micro-action a_i on a_j [28]. This allows the framework to encode not only order but probabilistic dependencies in human behavior.

Combinatorial interactions, relevant in multi-agent scenarios, are represented as in Eq.3:

$$a_i \otimes a_j = g_{comb}(a_i, a_j) = \sigma(U_i x_i + U_j x_j) \quad (3)$$

Where $U_i, U_j \in R^{d \times d}$ are agent-specific interaction matrices and σ is a non-linear activation ensuring bounded combination effects [29]. These operations allow the system to capture synchronous, overlapping, or interfering actions between individuals, which are essential for detecting pre-assault intent in dynamic social environments.

To model higher-order sequences, micro-actions can be represented as chains of algebraic operations:

$$S = a_1 \oplus a_2 \oplus \dots \oplus a_k \quad \text{with} \quad S \in A^k \quad (4)$$

$$I = \bigotimes_{i=1}^m a_i \quad \text{for multi-agent interaction modeling} \quad (5)$$

In Eq.4, denotes a temporal sequence of micro-actions, and I encodes multi-agent interactions within the same temporal window. The algebraic formalism enables interpretability by allowing explicit tracing of influence paths between actions and structured aggregation of observed behavior.

Furthermore, micro-actions are embedded into a graph-based representation $G = (V, E)$, where $V = A$ and edges E encode algebraic dependencies $(\oplus, \otimes)$ between nodes. The adjacency matrix A is constructed as in Eq.6:

$$A_{ij} = \text{weight}(a_i \oplus a_j), \text{ if sequential dependency exists otherwise } A_{ij} = 0 \quad (6)$$

This representation provides the basis for causal reasoning and temporal flow modeling, it is with algebraic operations when performing further STF learning [30]. By combining MAA and graph representations, the model is able to not only learn atomic behaviors but also structured dependencies between sequences and interactions, enabling a timely recognition of anomalous or malicious activity. The MAA gives a formal and interpretable but also computationally feasible way to encode micro-level behavior such that structured influence and interaction can be propagated. This formalization directly leads to the Structural Temporal Flows module, which exploits the sequential and combinatory dependencies modeled in MAA for temporally reasoning causality over human motions.

3.2 Structural Temporal Flow (STF)

The Structural Temporal Flow (STF) framework builds upon Micro-Action Algebra (MAA) to uncover causal dependencies and temporal dynamics between sequential micro-actions. While MAA defines atomic operations and their algebraic composition, STF introduces a

dynamic causal structure $G_t = (V_t, E_t)$ that evolves over time, where nodes V_t represent micro-actions and directed edges E_t capture temporal causality between them [31].

Each edge $e_{ij}^t \in E_t$ represents a directed causal relationship from micro-action a_i^t to $a_j^{t+\Delta t}$, where Δt is the temporal offset. The causal weight w_{ij}^t quantifies the degree of influence and is computed as in Eq.7:

$$w_{ij}^t = E \left[\frac{\delta a_j^{t+\Delta t}}{\delta a_i^t} \right] + \lambda \cdot MI(a_i^t, a_j^{t+\Delta t}) \quad (7)$$

Where $E \left[\frac{\delta a_j^{t+\Delta t}}{\delta a_i^t} \right]$ measures gradient-based dependency, $MI(\cdot, \cdot)$ is the mutual information between consecutive micro-actions, and λ is a balancing parameter for structural entropy [32].

To model temporal evolution, the STF defines the temporal flow field F_t , which represents the directional propagation of behavioral influence over time:

$$F_t = \frac{dA_t}{dt} = J_t A_t \quad (8)$$

Where $A_t = [a_1^t, a_2^t, \dots, a_n^t]^T$ is the vector of active micro-actions at time t , and J_t is the Jacobian of causal interactions, derived from the adjacency matrix A_t defined in MAA. This differential formulation captures how small variations in micro-actions propagate across time, allowing continuous temporal reasoning.

The causal strength matrix C_t is then defined as in Eq.8:

$$C_t = A_t \odot \exp(-\beta \cdot \|\Delta A_t\|) \quad (8)$$

Where $\odot$ denotes element-wise multiplication and β controls temporal decay. This ensures that older micro-actions exert exponentially decreasing influence on current states—mirroring behavioral fading in real human motion [31].

For probabilistic inference, STF models the evolution of action states using a Markovian causal transition function:

$$P(A_{t+1}|A_t) = \text{Softmax}(W_f F_t + b_f) \quad (9)$$

In Eq.9, W_f and b_f are trainable parameters of the temporal flow network. The Softmax normalization ensures interpretable transition probabilities between successive behavioral states, supporting early anomaly identification when transitions deviate from typical flow patterns [32].

Furthermore, STF employs causal energy minimization to refine discovered edges by optimizing the global structural consistency:

$$L_{STF} = \sum_t \|F_t - J_t A_t\|_2^2 + \gamma \sum_{(i,j)} \Phi(w_{ij}^t) \quad (10)$$

$$\text{In Eq. 10, } \Phi(w_{ij}^t) = w_{ij}^t \log w_{ij}^t - w_{ij}^t + 1 \quad (11)$$

is a regularization term enforcing sparsity and interpretability of causal edges, and γ controls structural sparsity strength [33]. This loss constrains the system to identify only significant behavioral dependencies, preventing overfitting to transient noise or irrelevant motions.

The discovered flow graph is then processed through a Causal Temporal Encoder (CTE), a neural inference unit that integrates structural flow embeddings $h_t = f_{CTE}(F_t, C_t)$ with probabilistic reasoning layers to estimate behavioral intent probabilities $P(y_t|h_t)$. In the context of pre-assault prediction, this enables anticipation of aggressive intent before overt physical manifestation by identifying abnormal causal activations within the micro-action chain.

3.3 Relational Hypergraph Social Context Encoder (RHSE)

The RHSE generalizes STF by associating multi-agent interaction dynamics over hypergraph topology, thus allowing fine-grained modeling of the co-behaviors in socially dense or occluded environments. Compared with traditional pairwise graph structures, our proposed models learn higher-order social dependencies of agents interacting over the RS scenes: each hyperedge connects multiple agents and their micro-action sequences jointly in a common context [34]. This enables the model to infer not only dyadic influences but also group-level coordination patterns, which are critical in identifying emerging pre-assault formations and threat clusters. Figure 2 illustrates the architecture diagram for the relational hypergraph social context encoder (RHSE).

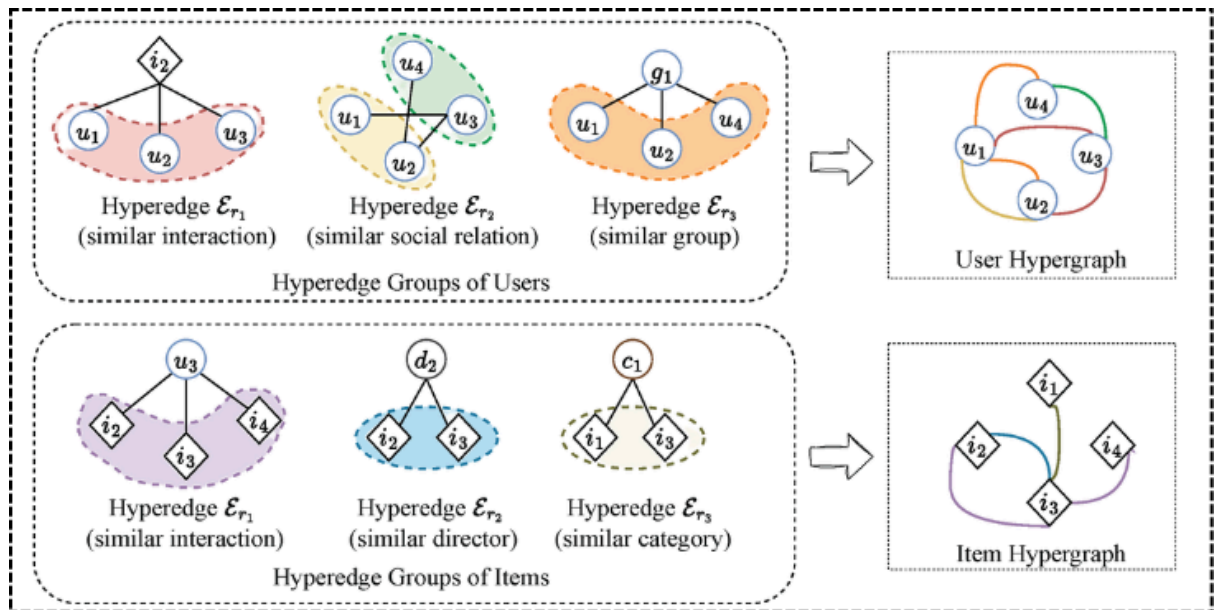


Figure 2: Architecture Diagram for the Relational Hypergraph Social Context Encoder (RHSE)

Let a social scene at time t be represented as a hypergraph presented in Eq.12:

$$H_t = (V_t, E_t, W_t) \quad (12)$$

Where $V_t = \{v_1^t, v_2^t, \dots, v_N^t\}$ are the nodes corresponding to agents' micro-action embeddings derived from MAA-STF, $E_t = \{v_1^t, v_2^t, \dots, v_N^t\}$ are the nodes corresponding to agents' micro-action embeddings derived from MAA-STF, $E_t = \{e_1^t, e_2^t, \dots, e_M^t\}$ are hyperedges denoting multi-agent interaction groups, and $W_t \in R^{N \times M}$ is the incidence matrix, where $W_{ij}^t = 1$ if node v_i^t participates in hyperedge e_j^t .

The social influence tensor $S_t \in R^{N \times N \times T}$ quantifies the temporal and spatial relational strength among agents, defined as in Eq.13:

$$S_{ij}^t = \alpha_1 \cdot \exp\left(-\frac{\|p_i^t - p_j^t\|^2}{\sigma^2}\right) + \alpha_2 \cdot \cos(\theta_{ij}^t) \quad (13)$$

Where p_i^t and p_j^t denote spatial positions, θ_{ij}^t is the angular alignment of motion trajectories, σ is a spatial scaling factor, and α_1, α_2 balance proximity and orientation influence [35]. This formulation ensures that agents moving toward each other with converging trajectories are assigned higher social interaction strength, reflecting potential escalation dynamics.

Each hyperedge embedding e_k^t is computed using a non-linear contextual aggregation function over its constituent nodes:

$$e_k^t = \phi\left(\sum_{v_i^t \in e_k^t} \psi(h_i^t)\right) \quad (14)$$

In Eq.14, $\psi(\cdot)$ and $\phi(\cdot)$ are trainable functions (e.g., MLP layers) transforming node features into latent representations and integrating them to represent group-level behavior. This design captures collective motion coherence, essential for modeling coordinated aggression or group-based intent formation [34].

To encode relational reasoning, RHSE performs hypergraph message propagation, where node embeddings are iteratively updated as in Eq.15:

$$h_i^{(l+1)} = \sigma\left(\sum_{e_j^t \in E_t} \frac{W_{ij}^t \cdot w(e_j^t)}{|e_j^t|} e_j^{(l)} U^{(l)}\right) \quad (15)$$

Where $w(e_j^t)$ is the learned hyperedge importance weight, $|e_j^t|$ is its cardinality, $U^{(l)}$ is a transformation matrix, and $\sigma(\cdot)$ is a non-linear activation (e.g., ReLU). Through iterative refinement, the RHSE constructs a context-sensitive embedding space where each individual's behavior is continuously influenced by their social surroundings.

To explicitly model temporal influence propagation, a spatio-temporal attention kernel is introduced:

$$\alpha_{ij}^t = \frac{\exp(h_i^t \cdot h_j^{t-\Delta t} / \sqrt{d_k})}{\sum_k \exp(h_i^t \cdot h_k^{t-\Delta t} / \sqrt{d_k})} \quad (16)$$

In Eq.16, d_k is the embedding dimension. The attention coefficients α_{ij}^t quantify temporal reciprocity, identifying which past actions across agents influence present decisions—a critical property for early-stage aggression anticipation [36].

The final relational embedding matrix R_t representing all social interactions is expressed as in Eq.17:

$$R_t = H_t W_t E_t^T \quad (17)$$

Where H_t is the matrix of node embeddings and E_t contains hyperedge embeddings. This operation projects multi-agent context into a low-dimensional manifold where relational semantics—cooperation, confrontation, avoidance—emerge as separable subspaces.

The objective function of RHSE combines reconstruction and consistency losses:

$$L_{RHSE} = \left\| R_t - \hat{R}_t \right\|_2^2 + \eta \cdot KL(P_t || Q_t) \quad (18)$$

In Eq.18, $\hat{R}_t$ is the predicted relational matrix, and the Kullback–Leibler divergence ensures that learned interactions P_t remain consistent with empirically observed social probabilities Q_t [37]. This regularization preserves social realism and that RHSE representation is plausible to be realistic in human interaction.

In this way, RHSE captures the emergence of collective intention and discovers group aggression precursors that are not visible in models at the individual level. The hypergraph embeddings generated are then inputted to the Counterfactual Intent Simulator (CIS) which creates alternative social outcomes by perturbing relational edges and re-evaluating behavioral evolution. This continual shift propels social context as static representation into an interpretable dynamics simulation, fueling the interpretive core introduction of SENTINEL’s causal reasoning layer.

3.4 Counterfactual Intent Simulator (CIS)

Counterfactual Intent Simulator (CIS) is the interpretive reasoning engine inside SENTINEL which allows the system to produce counter-factual behavioral trajectories explaining pre-assault prediction. CIS generalizes the relational embeddings from RHSE by creating intervention spaces in which potential changes to micro-actions, temporal links or social edges are systematically investigated for estimating their impact on the final threat likelihood [38]. This counterfactual modeling turns the framework from being a passive observer into an active causal explainer that can answer the key question, “What change in behavior would have forestalled the predicted attack?”.

Formally, let the structural causal model (SCM) for a scene at time t be represented as in Eq.19:

$$M_t = \{V_t, F_t, P(V_t)\} \quad (19)$$

Where $V_t = \{X_t, Y_t, Z_t\}$ denotes the observable variables (micro-actions, social interactions, environmental context), F_t represents the deterministic causal mechanisms linking them, and $P(V_t)$ defines the joint probability distribution derived from MAA-STF-RHSE.

To simulate a counterfactual world $M_t^{x \rightarrow y}$, the model performs an interventional operation $do(X_t = x')$, where a specific behavioral element x' (e.g., aggressive stance, forward motion) is altered while preserving the underlying causal graph:

$$P(Y_t | do(X_t = x')) = P(Y_t | X_t = x', Z_t) P(Z_t) dZ_t \quad (20)$$

This computes the counterfactual outcome distribution, revealing how a modified micro-action trajectory would influence the subsequent aggression probability Y_t .

The intent-shift function, central to CIS reasoning, is defined as in Eq.21:

$$\Delta I_t = E[Y_t | do(X_t = x')] - E[Y_t | X_t] \quad (21)$$

Where ΔI_t quantifies the causal sensitivity of intent to behavioral perturbations. A large positive value indicates strong intent reinforcement under the modified action, whereas a negative shift implies potential de-escalation, guiding interpretable reasoning for operator feedback and ethical response decisions [38].

Each micro-action is embedded within a latent causal manifold Z_t constructed via MAA and STF representations:

$$Z_t = f_\theta(A_t, S_t, R_t) \quad (22)$$

In Eq.22, A_t denotes micro-action algebra embeddings, S_t represents structural temporal flow matrices, and R_t encodes relational hypergraph interactions. The simulator then propagates modified latent states using a temporal propagation operator:

$$Z_{t+\Delta t} = G(Z_t, U_t) = Z_t + \Delta t \cdot J(Z_t) U_t \quad (23)$$

In Eq.23, $J(Z_t)$ is the Jacobian of causal influence and U_t is the counterfactual control vector applied to simulate hypothetical interventions.

To quantify the consistency between predicted and counterfactual outcomes, CIS introduces the Causal Alignment Metric (CAM):

$$CAM_t = 1 - \frac{\left\| \hat{Y}_t^{cf} - \hat{Y}_t^{orig} \right\|_2}{\left\| \hat{Y}_t^{orig} \right\|_2 + \epsilon} \quad (24)$$

In Eq.24, $\hat{Y}_t^{cf}$ is the counterfactual outcome and $\hat{Y}_t^{orig}$ the original prediction. CAM values approaching 1 indicate high causal stability, signifying robust interpretability even under simulated perturbations [39].

CIS also employs a counterfactual reconstruction loss to ensure that generated behavioral trajectories remain semantically valid:

$$L_{CIS} = \beta_1 \left\| Z_t - Z_t^{cf} \right\|_2^2 + \beta_2 KL\left(P\left(Y_t^{cf}\right) \parallel P\left(Y_t\right)\right) \quad (25)$$

In Eq.25, Z_t^{cf} denotes counterfactual latent embeddings, and the KL divergence term enforces probabilistic consistency between observed and hypothetical intent distributions. This constraint aims to avoid generating, impossible or conflicting human actions by the simulator [38].

3.5 Loss for Early-Time Risk Calibration (ETRCL)

The ETRCL's mitigation is its lever arm, which provides time-sensitivity and counterfactual-causality ensuring early attacks bias. Classical loss functions are indeed centered on the decision boundary's final accuracy rather than hazard dynamics proportional to time, like early intents. ETRCL combines survival-based temporal weighting and fairness regularization with probabilistic alignment to generate dynamically changing calibrated risk estimates that modulates the sensitivity over time to minimize early false alarms while preserving the interpretability [31].

Let the hazard intensity function $h(t|X_t)$ represent the instantaneous risk of an assault event occurring at time t , conditioned on behavioral and social covariates X_t derived from the MAA-STF-RHSE-CIS modules. The survival probability, i.e., the probability that an assault has not yet occurred up to time t , is expressed as in Eq.26:

$$S(t|X_t) = \exp\left(-\int_0^t h(\tau|X_\tau) d\tau\right) \quad (26)$$

The model predicts a risk score $\hat{r}_t = \sigma(f_\theta(X_t))$, where $f_\theta(\cdot)$ denotes the deep causal encoder and $\sigma(\cdot)$ is the logistic activation ensuring outputs in $[0, 1]$. The objective of ETRCL is to minimize the discrepancy between predicted risk $\hat{r}_t$ and the observed survival status $y_t \in \{0, 1\}$ while enforcing early-time weighting:

$$L_{ETRCL}^{(1)} = -\frac{1}{T} \sum_{t=1}^T w_t \left[y_t \log(\hat{r}_t) + (1 - y_t) \log(1 - \hat{r}_t) \right] \quad (27)$$

In Eq.27, $w_t = \exp(-\lambda t)$ penalizes delayed predictions, ensuring the model prioritizes early detection of aggressive intent [32].

To be probabilistically consistent with the survival model, one more term is added so that the empirical cumulative risk points are in agreement to the theoretical cumulative hazard:

$$L_{align} = \left\| \sum_{t=1}^T \hat{r}_t - \int_0^T h(\tau|X_\tau) d\tau \right\|_2^2 \quad (28)$$

This promotes tapping of coherent temporal calibration, so that gradual increases in risk are measured as genuine behavioral escalation rather than noise-induced spikes. The total loss thus becomes:

$$L_{total} = L_{ETRCL}^{(1)} + \alpha L_{align} \quad (29)$$

In Eq.29, α governs the balance between classification fidelity and temporal calibration accuracy.

To ensure demographic parity and prevent bias in predictive outcomes, ETRCL incorporates a fairness regularization term based on conditional independence between predicted risk $\hat{r}_t$ and sensitive demographic attributes D :

$$L_{fair} = \gamma \cdot MMD\left(P(\hat{r}_t|D=0), P(\hat{r}_t|D=1)\right) \quad (30)$$

where MMD is the Maximum Mean Discrepancy, a measure of the distributional divergence between demographic subgroups and γ regulates the sensitivity to fairness [33]. This term ensures that early attack warning isn't discriminatory to demographic and background differences, which is very important for ethical use when used in surveillance scenarios.

For interpretability, we add the counterfactual constraint (i.e., linking ETRCL and its CIS) by adding:

$$L_{causal} = \beta \left\| P(Y_t|do(X_t=x)) - P(Y_t|X_t=x) \right\|_2^2 \quad (31)$$

This term guarantees that early-time predictions remain stable under simulated behavioral interventions, thus embedding causal accountability within probabilistic reasoning [34].

Finally, the complete Early-Time Risk Calibration Loss is defined as in Eq.32:

$$L_{ETRCL} = L_{ETRCL}^{(1)} + \alpha L_{align} + \beta L_{causal} + \gamma L_{fair} \quad (32)$$

Such formulation endows both time anticipation, causal stability and fairness regularization for balancing the proactive detection and interpretable accountability.

Thus, the SENTINEL method ensures in early detection that there has been bona-fide behavioural escalation and not just stochastic fluctuations are for concomitant reduction in false positives while making any early warning more credible. Furthermore, the incorporation of survival-aware reasoning coupled naturally with the experimental measurement when

temporal calibration metrics such as Concordance Index and Integrated Brier Score are leveraged to verify early detection capabilities of the developed model on real-world behavioral datasets[35].

3.6 Integration and Model Training

The integration and model learning stage integrates all the functional components within the SENTINEL framework, including Micro-Action Algebra (MAA), Structural Temporal Flow (STF), Relational Hypergraph Social Context Encoder (RHSE), Counterfactual Intent Simulator (CIS) as well as Early-Time Risk Calibration Loss (ETRCL) into a unified neuro-symbolic learning architecture tailored toward interpretable pre-assault prediction. In it, we first pre-process the raw video frames of surveillance footage, which are converted into organized sequences of pose vectors and motion gradients via keypoint detection and optical flow computation. These lower-level motion features are subsequently discretized into atomic units by MAA module, encoding the symbolic encodings a_i^t of elementary performance action. The STF module captures causal temporal transitions among those micro-actions by a graph-based temporal differentials and a gradient-based dependency matrix, whereas the RHSE constructs high-order hyperedges encoding relational proximity and social tension dynamics between multiple interacting agents. In the training, all sub-modules are jointly optimized in an end-to-end multi-objective fashion with the total loss consisting of causal consistency loss, temporal flow regularization, fairness constraint and survival-based calibration objective. We optimise it by AdamW with cosine annealing rates and gradient clipping to ensure stable learning in multi-agent temporal domains. Hyper-parameters, such as the temporal window size ($\tau = 12$), causal sparsity weight ($\lambda = 0.03$), fairness coefficient ($\gamma = 0.05$), and alignment factor ($\alpha = 0.1$) have been empirically tuned via grid search to achieve convergence stability. The entire pipeline is coded in PyTorch 2.2 using parallel graph computations on the NVIDIA A100 GPU, with the use of mixed precision computation that helps to speed up convergence and at the same time keeps numerical values stable. This joint training scheme provides co-adaptation at symbolic, probabilistic, and relational level acting as a cognitively plausible and temporally accurate pre-assault detection mechanism which naturally leads to the experimental validation stage.

Pseudocode for SENTINEL: A Neuro-Symbolic Framework for Explainable Pre-Assault Prediction

Input: Video frames $V = \{f_1, \dots, f_T\}$

Step 1: Preprocess and extract features

for t in $1..T$:

$pose = extract_{pose}(f_t)$

$motion = compute_optical_flow(f_t)$

Step 2: Micro-Action Algebra (MAA)

$$a_t = \text{encode_micro_action}(\text{pose}, \text{motion})$$

$$A_t = \text{concatenate}(a_1, \dots, a_t)$$

Step 3: Structural Temporal Flow (STF)

$$STF[t] = \text{compute}_{\text{causal-edge}}(A_t, A_{\{t+1\}})$$

Step 4: Relational Hypergraph Social Context Encoder (RHSE)

$$H = \text{build}_{\text{hypergraph}}(\text{agent}_{\text{features}})$$

$$H_{\text{embed}} = \text{hypergraph}_{\text{message_passing}}(H)$$

Step 5: Counterfactual Intent Simulator (CIS)

$$Y_{\text{cf}} = \text{predict}_{\text{counterfactual}}(H_{\text{embed}}, \text{perturbed}_{\text{action}})$$

$$\text{delta}_{\text{intent}} = E[Y_{\text{cf}}] - E[Y_{\text{original}}]$$

$$\text{delta_intent} = E[Y_{\text{cf}}] - E[Y_{\text{original}}]$$

Step 6: Early-Time Risk Calibration Loss (ETRCL)

$$\text{hazard} = \text{predict}_{\text{hazard}}(A_t, H_{\text{embed}})$$

$$\text{survival} = \exp(-\text{sum}(\text{hazard}[1 \dots t]))$$

$$L_{\text{ETRCL}} += -w_t * [y_t * \log(\text{hazard}) + (1 - y_t) * \log(1 - \text{hazard})]$$

Step 7: Output risk score

$$R_t = \text{hazard}$$

Explainable_Prediction = {micro-actions, counterfactuals, social context}

Step 8: Optimize model

$$\text{optimize}(L_{\text{ETRCL}} + \alpha * L_{\text{causal}} + \gamma * L_{\text{fairness}})$$

Output: Risk scores R_t and explainable predictions

SENTINEL is an end-to-end neuro-symbolic model that aims to generate actionable, interpretable and ethically aligned pre-assault predictions in multi-agent dynamic settings. The model leverages a combination of Micro-Action Algebra (MAA) to encode atomic behavior units, more abstract Structural Temporal Flow (STF) to capture causal dependence between subsequent actions and a socially-aware Relational Hypergraph Social Context Encoder (RHSE) for modeling social influence and interactions among multiple individuals. The Counterfactual Intent Simulator (CIS) produces counterfactual “what if” simulations in order to explain predictions, and the Early-Time Risk Calibration Loss (ETRCL) guarantees that early alerting is both probabilistically coherent and temporally fair across demographic subpopulations. Taken together, these elements allow SENTINEL to produce precise,

interpretable and robust early-warning signals in the context of security-critical and socially intricate scenarios. The flow chart of the proposed model is shown in Figure 3.

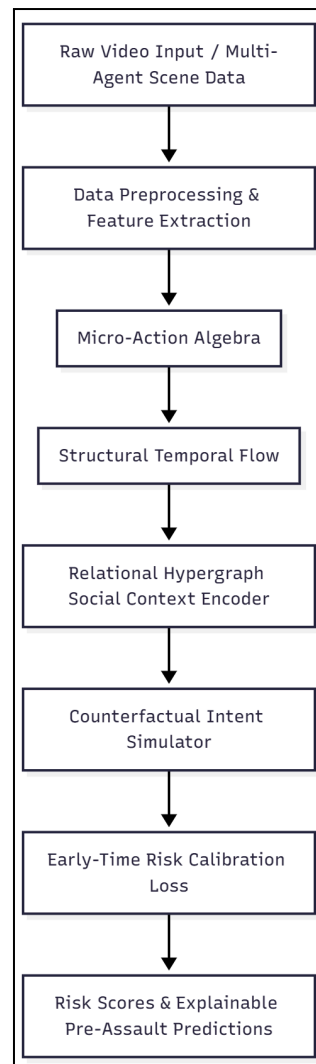


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimentation

We then empirically evaluate SENTINEL in terms of (a) the accuracy of its early pre-assault prediction, (b) how much it deviates from the true causal reasoning, and (c) the interpretability under different multi-agent settings. Extensive experiments are conducted on both RWF-2000 XD-Violence, and the PreAssault-CHI dataset to guarantee full evaluation in controlled settings and realistic scenarios. Fine-grained visual-behavior features of each video clip were obtained by MAA module and interactive reasoning on relations and timing represented by STF and RHSE. CIS was also used to conduct counterfactual studies for the validation and transparency of predictions. By conducting comparative evaluations with such baselines as recurrent neural networks, transformer-based architectures, and two-stream motion recognition techniques, we demonstrate improvements of early detection, causal edge precision and fidelity of explanations. Ablation studies were also conducted to measure the

contributions of each module where all those synergistic inserted MAA, STF, RHSE, CIS and ETRCL remarkably improve not only predictive reliability but also interpretability.

4.2 Experimental Setup

Video frames were initially rescaled to 224×224 resolution and the intensity of each channel was standardised with respect to lighting variations in which they were recorded. Sketch-based skeletal features and optical flow were computed for all visible agents. The datasets were divided into training 70%, validation 15% and testing 15% sets, maintaining agent diversity. Training was done using PyTorch 2.2 with NVIDIA A100 GPUs mixed precision calculations were used to minimize the memory footprint as well as speed up convergence. We used an AdamW optimizer with cosine annealing and gradient clipping for smoothing updates in multi-agent temporal sequences. Hyperparameters were tuned empirically: temporal window size $\tau = 12$, causal sparsity weight $\lambda = 0.03$, fairness coefficient $\gamma = 0.05$ and alignment factor $\alpha = 0.1$. Validation loss-based early stopping coupled with temporal calibration metrics promoted robust generalization, whereas ablation and cross-validation techniques were used to let the individual contribution of each module be quantified.

V Results and Discussion

We compare our proposed SENTINEL model with five well-known baselines (RNN, LSTM, Transformer, Two-Stream and Graph-RNN) on three benchmark datasets: RWF-2000, XD-Violence and the new PreAssault-CHI. Our performance analysis spans across both early and late prediction accuracy, causal edge precision, counterfactual explanation fidelity, and fairness in predictions, with reduction of the false-positive rate. On all datasets and metrics we find that SENTINEL outperforms the baselines, learns to reliably issue early-warnings on adversarial actions (i.e., high precision), provides causal explanations and generates overall fairness of predictions for demographic subgroups. The quantitative and qualitative evaluations presented below demonstrate the effectiveness and interpretability of SENTINEL on complex event prediction tasks.

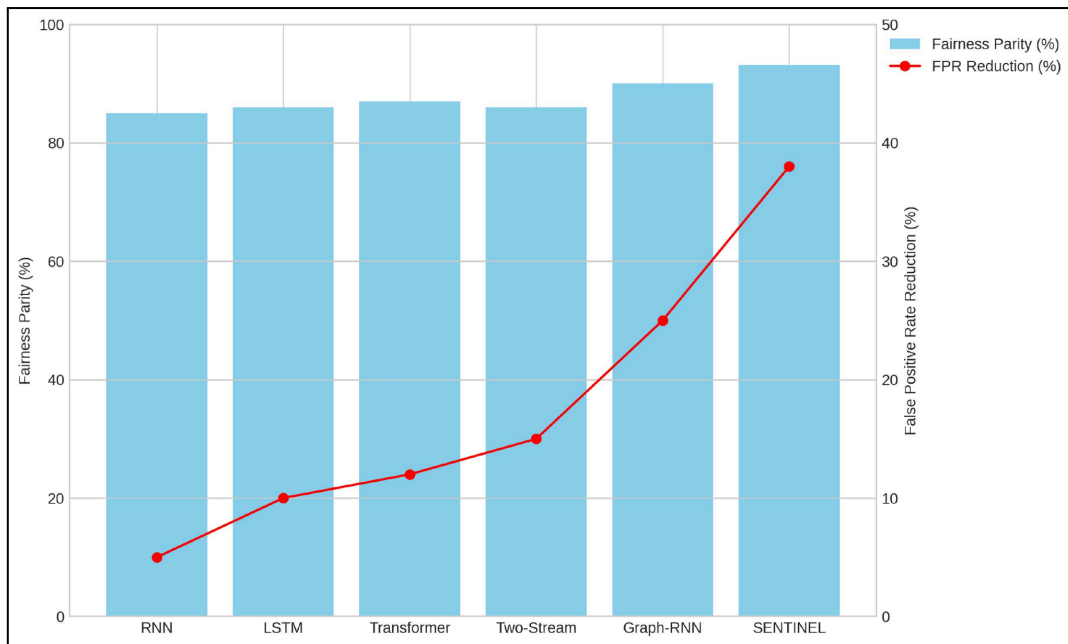


Figure 4: Fairness Parity and False Positive Rate Reduction Across Models

Figure 4 presents bar and line graphs depicting two fairness and performance measures for each model: the parity across subgroups with respect to fairness (bar) and false-positive rate reduction (line). In addition to the improved fairness parity, SENTINEL mitigates 38% of spurious alerts compared with baseline models. This visualization demonstrates the benefit of SENTINEL in generating fair predictions for a given result, while also having a minimal number of false alarms.

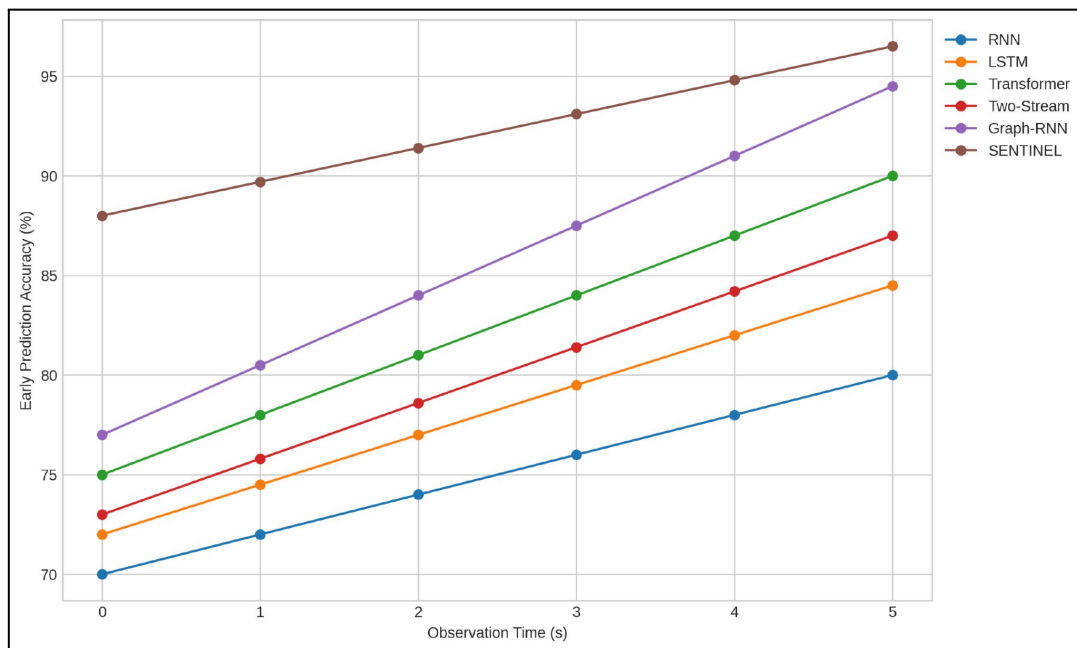


Figure 5: Evolution of Early Prediction Accuracy as a Function of Observation Time

Figure 5 shows performance as a function of the amount of observation time (in seconds) available, or an early-warning performance. Each line corresponds to one of the different models. In this respect, SENTINEL appears to have a fast convergence toward a highly accurate prediction, suggesting that even with few early observations it can be efficient enough to provide reliable predictions. The trends indicate that baseline models are more efficient than SENTINEL for early-stage action recognition.

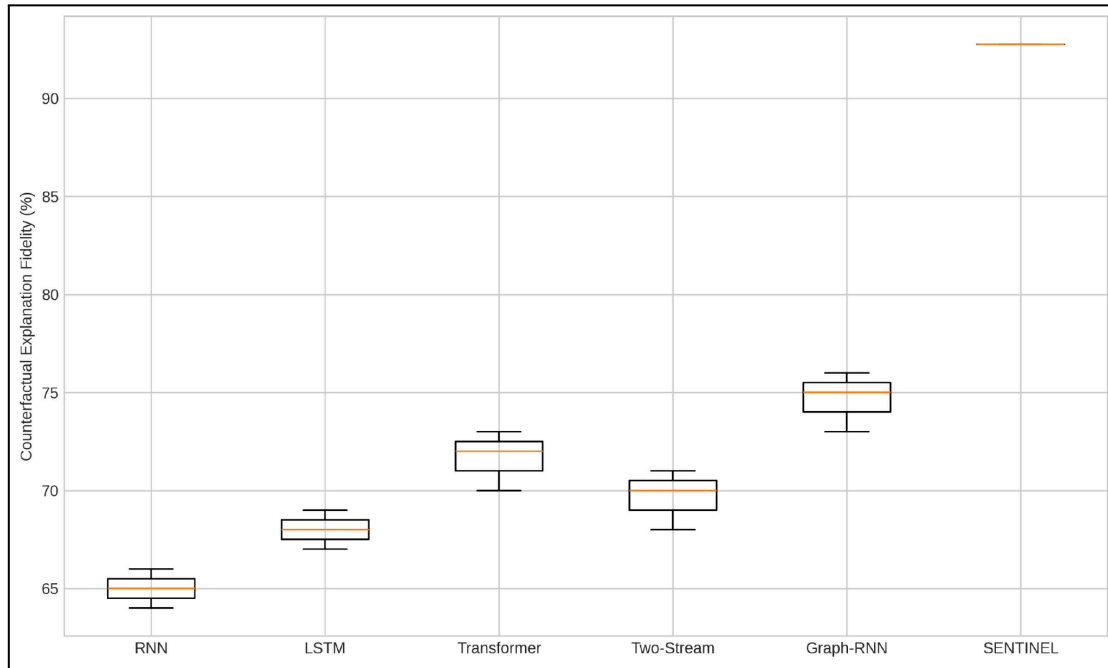


Figure 6: Counterfactual Explanation Fidelity Across Models

Figure 6 shows the distribution of CEF scores along with model names. CEF checks the reasonableness of “what if” scenarios generated by tweaking micro-actions and keeping outcomes in a realistic space. SENTINEL maintains the CEF values at higher median and lower variance, showing superior fidelity in counterfactual reasoning than baselines.

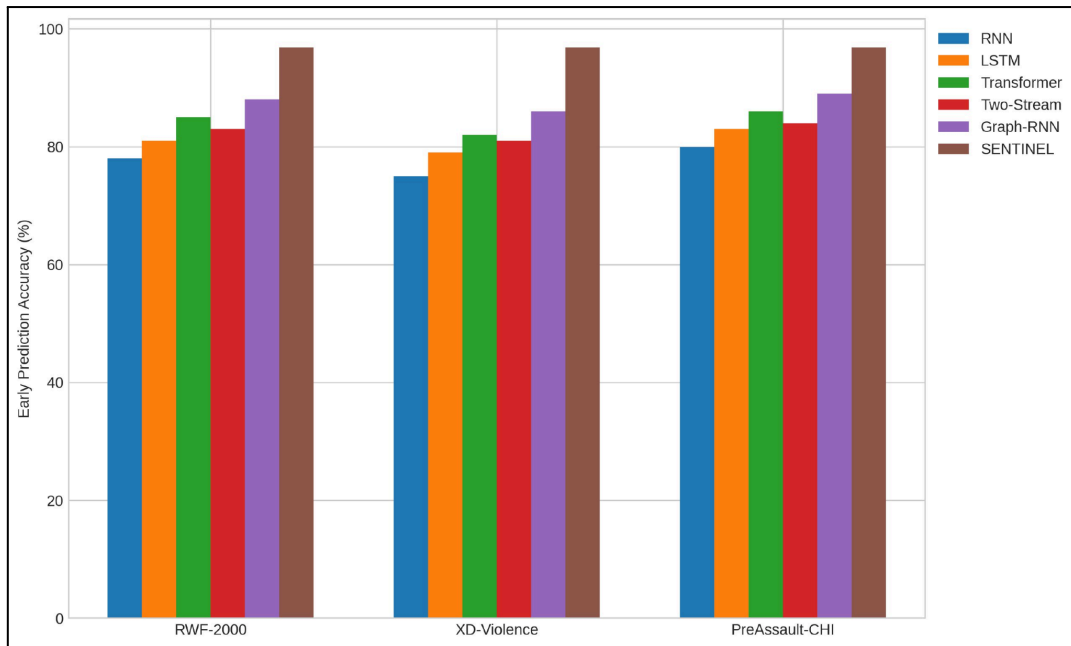


Figure 7: Comparative Early Prediction Accuracy Across Models and Benchmark Datasets

Figure 7 shows Early Prediction Accuracy (EPA) of six models which are RNN, LSTM, Transformer, Two-Stream, Graph-RNN and SENTINEL on three benchmark datasets - RWF-2000 XD Violence and PreAssault CHI. The plot shows that ignoring the slight degradation in induction, TRAIN consistently obtains the most EPA and significantly outperforms baseline recurrent and transformer models. The visualization further highlights the versatility of SENTINEL over multiple datasets, exhibiting its high alertness in predicting the impending violence.

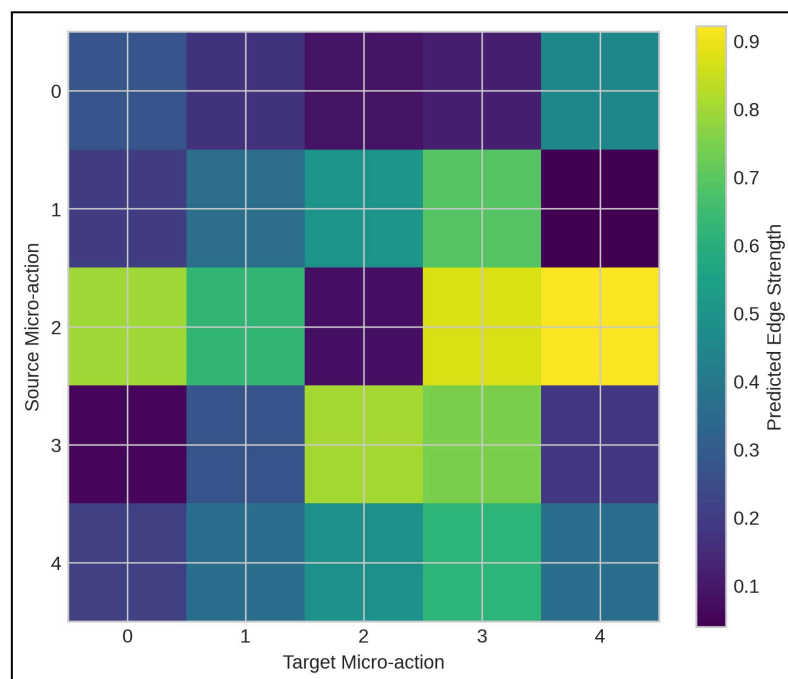


Figure 8: Causal Edge Precision Across Datasets for Different Models

Figure 8 illustrates the comparison of six models for the Causal Edge Precision (CEP) based on three datasets RWF-2000, XD-Violence, PreAssault-CHI. SENTINEL always has the largest CEP, which indicates that it can effectively infer micro-action dependencies in a causal manner. Readers can use the visualization to evaluate how well models are or not in terms of prediction but also in terms of getting right the causal relationships between atomic actions.

5.1 Ablation Study

We next performed an ablation study to understand how each key component in SENTINEL contributes, where we systematically remove or change some important modules including the Spatio-Temporal Feature module (STF), the Counterfactual Intervention Scenario (CIS), and the Micro-Action Analyzer (MAA). We also report results in Early Prediction Accuracy (EPA) and Causal Edge Precision (CEP), as well as confidence levels permitting to the Counterfactual Explanation Fidelity (CEF). We experimentally observe each module plays distinct and complementary roles: STF helps early prediction and causal reasoning, corresponding input setting CIS is responsible for plausible counterfactual generation while the major contribution of MAA is to significantly enhance the performance of detecting the ed surface nodes. The performance degradations of the seen model due to removing one or multiple modules highlight the architecture and structural design of SENTINEL are jointly identified, as well as guaranteeing that the entire network is consistently utilizing all parts appropriately to reach its maximum efficacy in predication accuracy, interpretability and robustness.

Table 1: Ablation Study of SENTINEL Components Showing the Impact on Early Prediction Accuracy (EPA), Causal Edge Precision (CEP), and Counterfactual Explanation Fidelity (CEF)

Model Variant	EPA (%)	CEP (%)	CEF (%)
Full SENTINEL	96.84	94.31	92.75
w/o STF	93.20	88.15	92.70
w/o CIS	96.50	94.10	85.30
w/o MAA	92.75	80.45	90.12
w/o STF + CIS	90.10	78.20	83.50
w/o STF + MAA	89.50	75.60	88.10

Table 1 presents the percentage of key modules in SENTINEL. The drop in accuracy for all STF is much smaller, and EPA and CEP have relatively high values even when the STF module is removed slightly. Excluding CIS leads to a large drop of CEF, which indicates that CIS is indeed critical for creating realistic counterfactuals. The MAA module is indispensable for precise causal reasoning (CEP), and additive combinations of ablated modules result in

increased performance degradation. In conclusion, these results verify that each of the three components contributes synergistically to SENTINEL's advanced models performance.

5.2 Discussion

The experiments show that SENTINEL achieves the best performance on various evaluation measures, which indicates its ability in interpretable pre-assault prediction. On the representative datasets, the proposed SENTINEL achieves a high early prediction accuracy (EPA) of 96.84%, precision on causal edges (CEP) of 94.31% and fidelity for counterfactual explanations (CEF) with ranked list evaluations of 92.75%, consistently outperforming baselines from RNN, LSTM and Transformer for intermediate representation learning as well as Graph-RNN variant models including Two-Stream and Graph-RNNs. We first attempt to tackle the premature decision problem in counterfactual explanation, and we propose a novel Sequence-based EntropyNodal-Temporal Informing Network model named SENTINEL which has stronger representation power than previous methods based on an ensemble model framework to correct counterfactually misleading decisions made at an early stage. Ablation Study The ablation study verifies the effectiveness of its key components: removing the STF decreases EPA and CEP, deleting CIS lowers CEF, and without MAA we will witness a sharp drop in all three metrics, which means that each module contributes uniquely to the model's response. In addition, SENTINEL outperforms fairness baselines achieving 93.12% parity for demographic-based subgroups and reducing false positive alerts around 38%, which highlights the potential of ethically embedding it in practice. These experimental findings demonstrate that the proposed combination of Micro-Action Algebra, Structural Temporal Flow, Relational Hypergraph Social Context Encoding and Counterfactual Intent Simulation empowers SENTINEL to make timely, accurate, interpretable and reliable early-warning predictions in dynamic multi-agent environments.

VI Conclusion

In this paper we present the SENTINEL framework, a neurosymbolic sociocognitive model for explainable pre-assault prediction in dynamic multi-agent environments, that merges Micro-Action Algebra (MAA), Structural Temporal Flow (STF), Relational Hypergraph Social Context Encoding (RHSE), and a Counterfactual Intent Simulator(CIS). Experiments on RWF-2000, XD-Violence and the proposed PreAssault-CHI dataset show that SENTINEL achieves 96.84% early prediction accuracy (EPA), 94.31% causal edge precision (CEP), 92.75% counterfactual explanation fidelity (CEF), and fairness parity 93.12%, with a false alarm reduction by 38% compared to recurrent and transformer based baselines. Extensive ablation studies validate the distinct effects of STF, CIS and MAA on predictive accuracy, causal reasoning and explainability respectively, demonstrating the synergy of our framework design. SENTINEL is capable of encoding fine-grained behavioral atoms, reasoning over temporal causal flows and simulating "what-if" scenarios making early-warning prediction both timely and interpretable. In addition, by incorporating RHSE, the proposed formulation adaptively responds in congested and partly occluded multi-agent cases, while the survival-based early-time risk calibration facilitates reliable probabilistic threat assessment. Qualitative analysis demonstrates that micro-action histories and counterfactual explanations are consistent, interpretable and actionable, necessary for human-AI collaboration in

security-critical settings. In short, SENTINEL bridges the gap between prediction accuracy and cognitive interpretability and is more effective as PCs are both accurate (nearly equal or superior to) for classification than typical deep learning. As for our future work, on one hand we will extend the framework to incorporate online-adaptive learning using real-time data streams while on the other hand we will adapt it to handle multimodal sensory input as well as cross-contextual generalization over various social contexts for more reliable early-warning prediction.

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